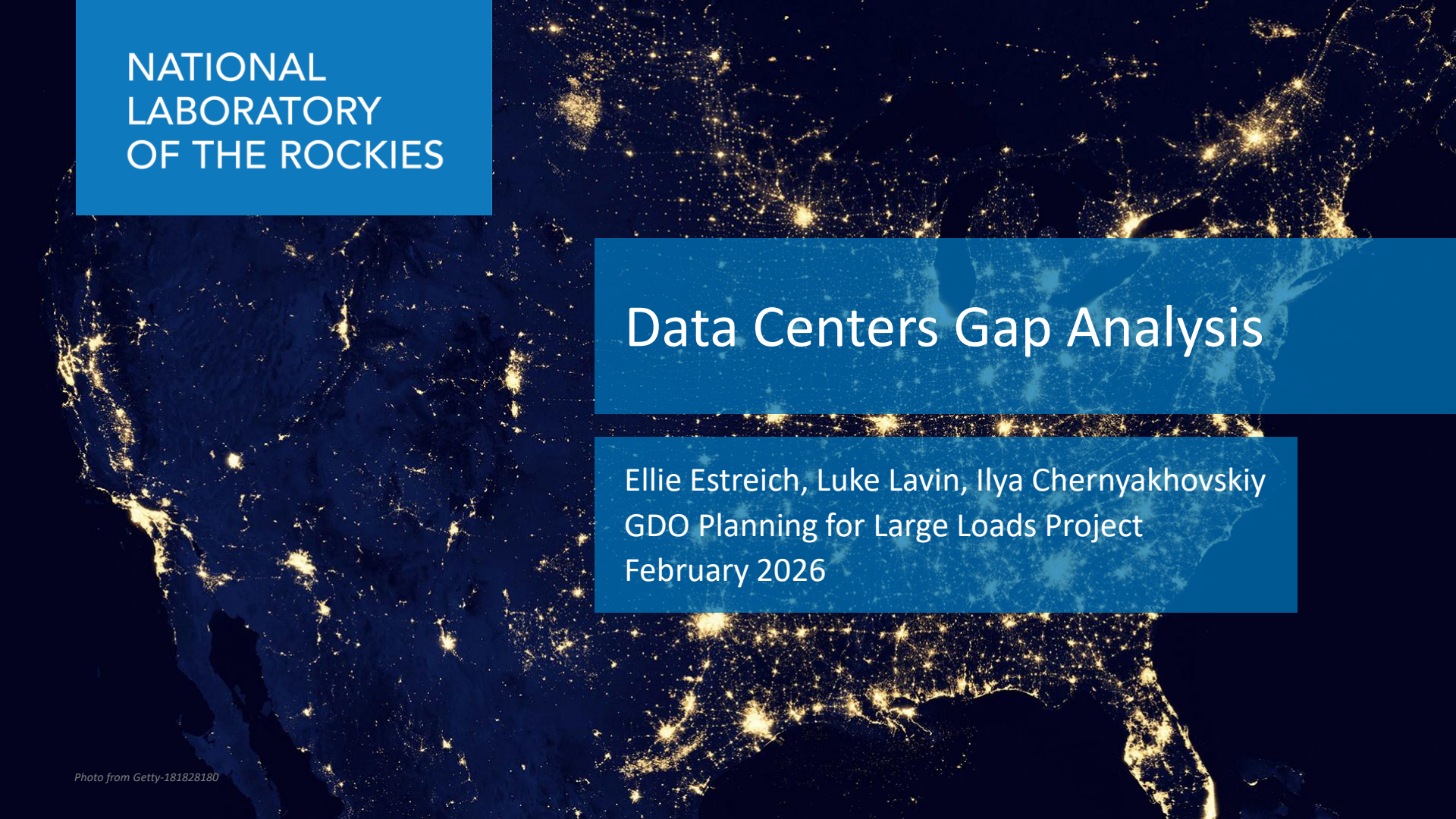




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Data Centers Gap Analysis

Ellie Estreich, Luke Lavin, Ilya Chernyakhovskiy
GDO Planning for Large Loads Project
February 2026

Data centers gap analysis

- **We analyze gaps in understanding** U.S. grid and customer impacts of large load development.
- **We also identify and categorize mitigation options** for near-term large load integration with the U.S. bulk power system.
- **Research questions:**
 - How are large loads driving near-term electricity growth?
 - Which factors are important for siting large loads?
 - What are realistic near-term mitigation options?
 - Where are gaps in the data that would allow us to understand and/or model large loads better?
 - Where should NLR prioritize adding value?





1 Executive Summary

2 Large Load Demand Growth

3 Mitigation Options for Large Load Integration

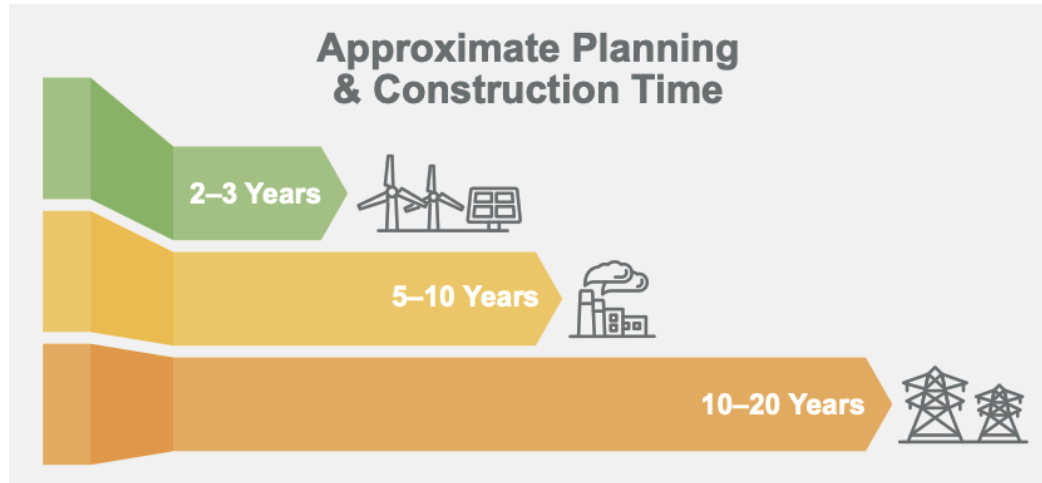
4 Options for Labs To Advance Large Loads Modeling

5 Appendix

Executive Summary

Large loads expected to drive near-term electricity demand growth, but with significant uncertainty

Example of planning and construction timelines for new generation, load, and transmission & distribution infrastructure:

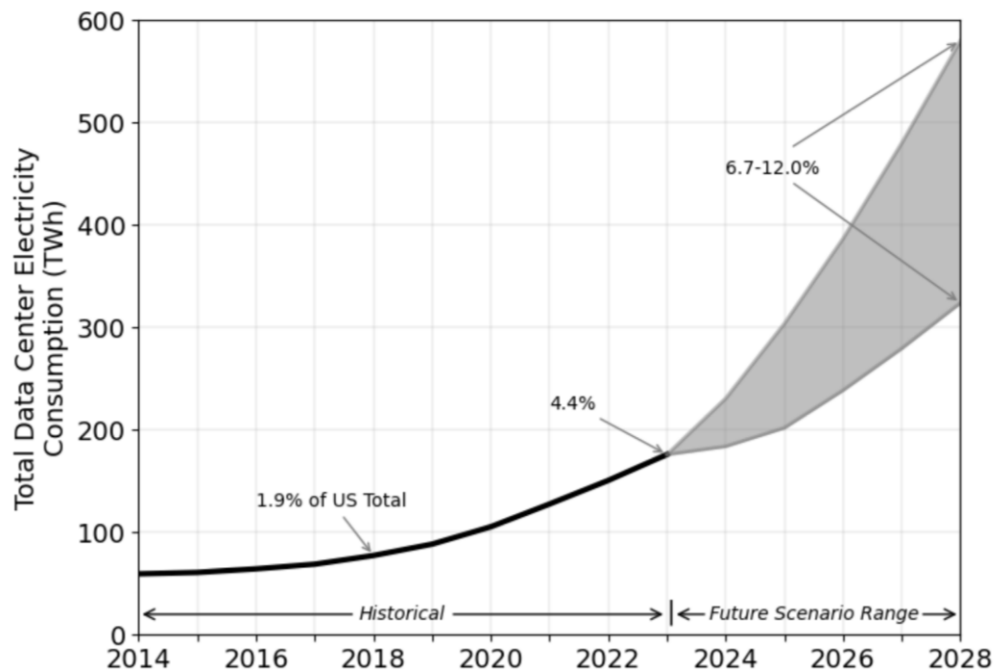


Source: Lee et al.

Problem: Historically, it has taken 5+ years to plan, permit, site, and build grid infrastructure.

How can we reduce uncertainty for long-term investments with potentially huge grid impacts?

Large loads electricity growth forecasts are subject to considerable uncertainty



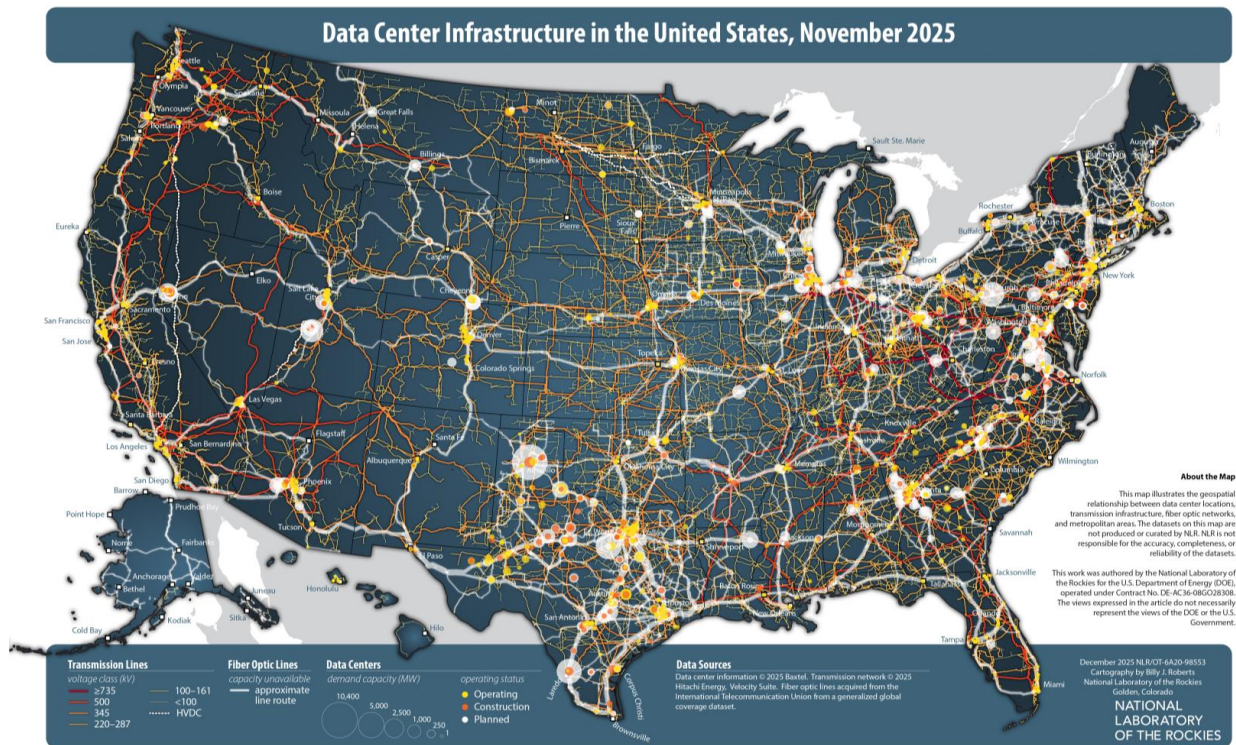
Total U.S. data center electricity use from 2014 through 2028;

Source: Shehabi et al., 2024.

Resolving uncertainty: How can we track which future we are headed toward and course-correct grid planning?

How and where can a capacity-strained grid quickly and flexibly add tens to hundreds of gigawatts of new load?

Data and interactive map produced under the U.S. Department of Energy's (DOE's) [Speed to Power Initiative](#)



Different mitigation options and emerging practices potentially improve speed to power

“HEDGE”: Tariffs



How? Structure tariffs to encourage demand flexibility or other investments.

Impact: Ensure large loads pay incremental costs.

Examples: Demand flexibility, storage, rates, and other tariff provisions

“ALIGN”: Streamlining Interconnection



How? Fast-track solutions to more efficiently use the grid and bring on new supply.

Impact: Align timelines for supply with large loads.

Examples: Grid-enhancing technologies, provisional interconnection

“MATCH”: Co-location



How? Large loads physically locate and financially contract with proximate generation.

Impact: New large loads bring own matching supply.

Examples: On-site generation, front-of-meter arrangements, islanded arrangements

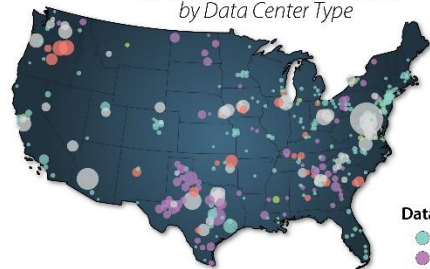
What metrics can help reduce uncertainty around data center buildout?

Data Center Markets, 2025

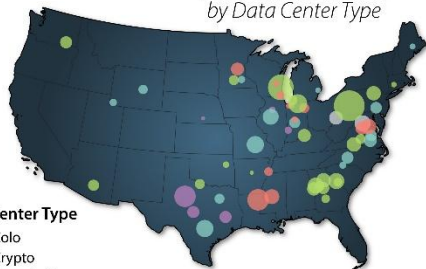
*All Data Center Markets
by Market Category*



*Existing & Emerging Markets
by Data Center Type*



*Prospective Markets
by Data Center Type*



Data Center Type

- Colo
- Crypto
- Greenfield
- Hyperscale
- Multiple

About the Data

Transmission network and data center information © 2023 Hitachi Energy.
Velocity Suite. Fiber optic lines were acquired from the International
Telecommunication Union from a generalized global coverage dataset.

August 2025
Cartography by Dr. J. Roberts

Why are certain markets emerging and not others?

- **Metrics to track uncertainty** could have a near-term impact on decision-making.
- **Independent, unbiased information** can help regulators make real-time decisions.
- **Quantifiable components:**
 - **Favorability:** Track megawatts (MW) by market and large load type; output metrics on which areas/criteria have appeared favorable
 - **Grid costs:** Bulk system cost projection scenarios
 - **Hedging:** Track in-use large load tariffs
 - **Aligning:** Track prevalence of co-location arrangements and types
 - **Matching:** Track interconnection processes
 - **Mitigating:** Longer-term options, system headroom and grid-enhancing technologies (GETs)

Tracking key data center growth uncertainty metrics requires developing new datasets and modeling

Possible steps for **national laboratories to improve data center representation and siting** analysis include:

1

Develop **representative load profiles** for different data center types

2

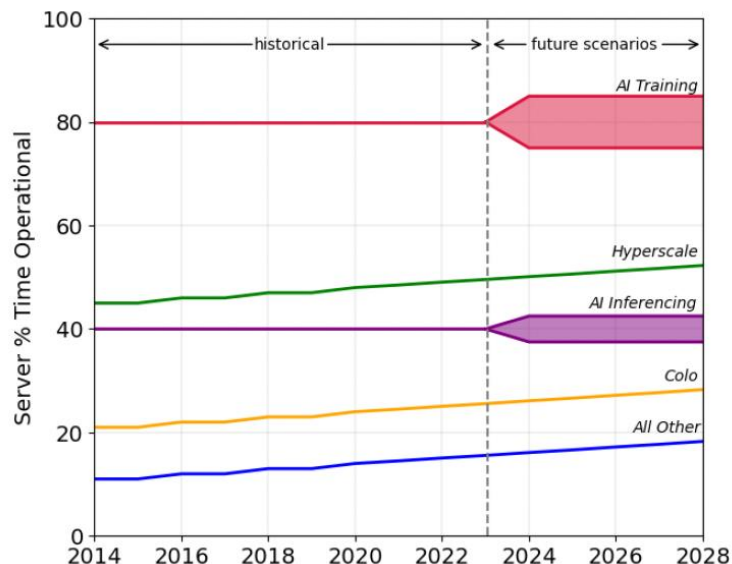
Track metrics on **co-location arrangements** to improve modeling representation of co-location options

3

Track detailed data on **in-use large load tariffs**

1. Develop representative data center load profiles

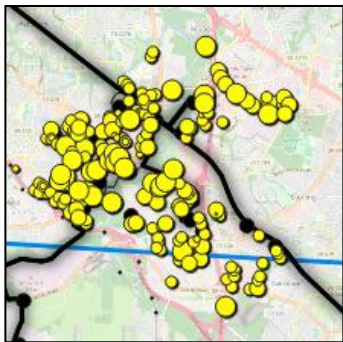
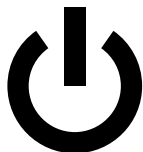
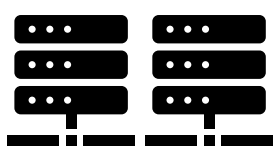
Operational time of servers given data center type (Shehabi et al., 2024)¹



- **Capture time series of characteristic AI workloads** by developing representative load profiles (i.e., representing a typical pattern of consumption over a specified time interval).
 - Develop profiles at sufficient temporal and geographic resolution for common data center categories.
 - Can capture other unique characteristics, e.g., utilization of rated demand capacity.
- **Developing canonical load profiles enables better answers to key questions:**
 - What are achievable levels of demand flexibility from data centers?
 - What mix of generation and on-site resources cost-effectively serves growing data center demand?

¹Shehabi, A., et al. 2024. 2024 United States Data Center Energy Usage Report. Lawrence Berkeley National Laboratory. LBNL-2001637. <http://dx.doi.org/10.71468/P1WC7Q>.

2. Track co-location arrangements



Data Center Capacity (MW)

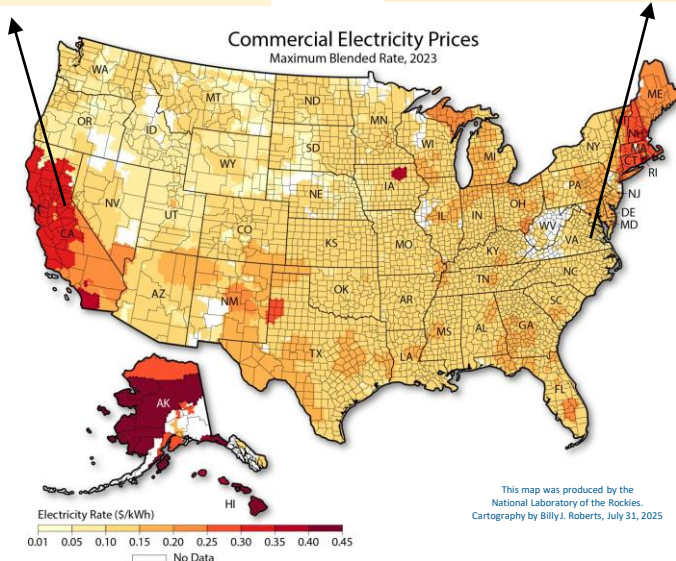
- 0 - 1.6
- 1.6 - 8
- 8 - 30
- 30 - 50
- 50 - 1000

- **Tracking metrics on co-location arrangements** could help address siting questions
 - Co-location types (e.g., behind- vs. front-of-meter, single/shared point of interconnection (POI), fully islanded, etc.)
 - Speed to power.
- **Tracking co-location options enables better answers to key questions:**
 - Which co-location arrangements are improving speed to power?
 - Does co-location provide contingency benefits to the grid?
 - How can co-location reduce costs of grid upgrades?
 - Where can co-location create additional siting opportunities for data centers?

3. Track in-use large load tariffs

California SB57 requires establishment of specific rate structures for data centers.

Virginia SCC approves new rate class for large customers



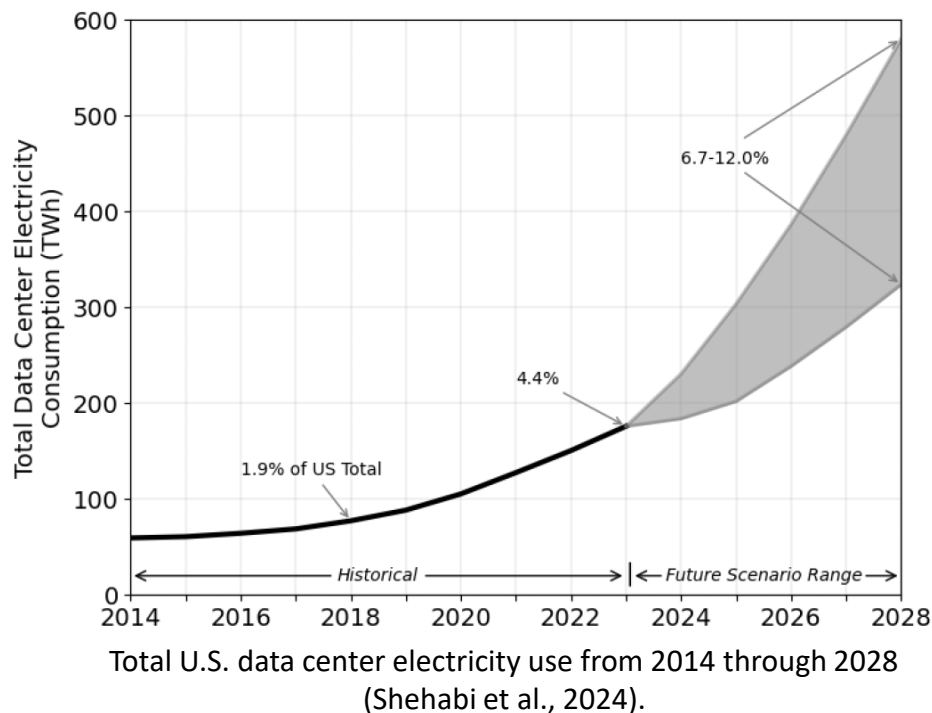
Source: Billy Roberts, NLR 2025.

- **Develop a public, NLR-vetted detailed dataset of large load tariffs** could help researchers evaluate demand flexibility, cost impacts, and more.
- **NLR already has a framework/database** (Utility Rate Database)¹ and AI tools to advance this work.
- **Tracking in-use large load tariffs enables better answers to key questions:**
 - How are different regions and utilities handling cost allocation for large loads?
 - Do we observe load flexibility resulting from tariff choices?
 - Do we observe data centers using on-site generation to avoid curtailment or high prices from applicable tariffs?

¹Zimmy-Schmitt, Daniel, and Jay Huggins. *Utility Rate Database (URDB)*. NLR, June 1, 2010. Distributed by Open Energy Data Initiative (OEDI). <https://data.openei.org/submissions/5>.

Large Load Demand Growth

Data centers driving near-term electricity demand growth



- Investments in data centers and other large loads are a driver for **unprecedented, near-term accelerated load growth**.
 - Recent LBNL report suggests data centers could account for 7% to 12% of all U.S. electricity consumption by 2028.¹
 - Data center development is “clustered”² in specific markets and geographies.
- Demand projections are subject to considerable uncertainty with differences across current projections.^{1,3}
 - Low visibility across load interconnection queues and “phantom requests”
 - Future uncertainty with respect to chips, input costs, energy efficiency improvements, rates, etc.

¹Shehabi, A., et al. 2024. 2024 United States Data Center Energy Usage Report. Lawrence Berkeley National Laboratory. LBNL-2001637. <http://dx.doi.org/10.71468/P1WC7Q>.

²Powering Data Centers. 2024. EPRI. <https://www.epri.com/research/products/000000003002031198>.

³Patel, K., et al. 2024. Virginia Data Center Study. E3. https://jlarc.virginia.gov/pdfs/presentations/JLARC%20Virginia%20Data%20Center%20Study_FINAL_12-09-2024.pdf.

High-demand growth driving interest in solutions to improve speed-to-power



Image by Joe DelNero, National Laboratory of the Rockies 101346

- **Against the backdrop of a capacity-strained grid, developers are seeking faster timelines** than it typically takes to add new capacity and accommodate new load additions on the grid.
- **Data centers also tend to have unique characteristics** that may impact possible mitigation solutions.
 - For example, data centers express high willingness to pay for electricity, which may constrain opportunities for shifting energy in absence of effective incentives.¹
- **The potential supply/demand mismatch** is creating reliability and resiliency risks and driving interest in faster-track solutions.

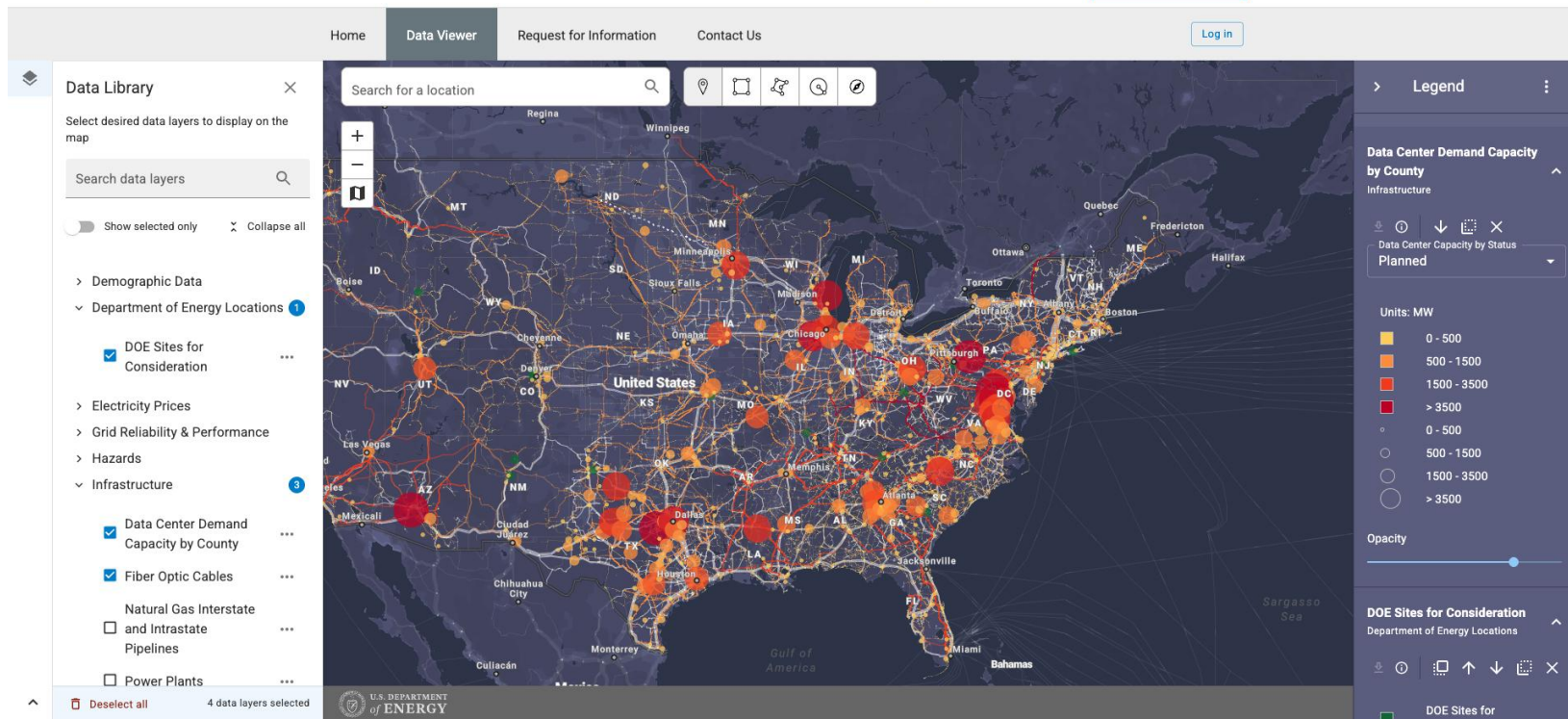
¹"Can Data Centers Flex Their Power Demand?" 2025. Energy Institute at Haas (blog). April 14, 2025.
<https://energyathaas.wordpress.com/2025/04/14/can-data-centers-flex-their-power-demand/>.

Speed to Power initiative visualizes existing infrastructure and known data center locations

Accelerating Speed to Power



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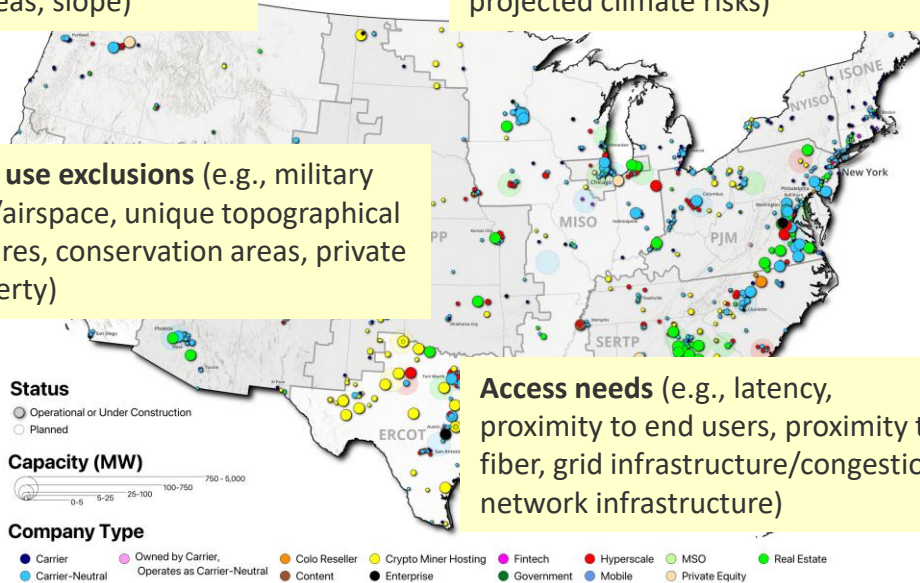
Siting tool captures geospatial factors affecting future data center development

Land use development
(e.g., wetlands, forests, urban areas, slope)

Environmental/climate (e.g., water resources, ambient humidity, projected climate risks)

Land use exclusions (e.g., military land/airspace, unique topographical features, conservation areas, private property)

Access needs (e.g., latency, proximity to end users, proximity to fiber, grid infrastructure/congestion, network infrastructure)



- **Data center siting and development** depends on a range of environmental, economic, and/or policy-related factors.
- **Categories of geospatial factors are at left;** NLR has already completed work on this.
- **Latency**—average time for data to travel between the data center and user—is an important siting factor.
 - Developers seek to minimize latency.
 - Latency depends on several geospatial factors (e.g., network infrastructure and proximity to end users).

Source: U.S. Data Center Sites. Baxtel, 2025. Accessed June 16, 2025. <https://baxtel.com>.

Some factors are more difficult to capture in nationally representative data but impact siting decisions

- **Vacancy rates**¹—e.g., investors may be more confident financing projects in top-tier markets with low vacancy rates
- **Electricity rates and grid costs**, supply chain issues, and other input costs and trends
- **Lessons from Northern Virginia, Texas, and other regions with clustered data center development** show that qualitative factors have the potential to impact investment and siting decisions.
 - **Zoning and permitting laws**
 - **Local and state tax laws and incentives**²—tax breaks for data center developers
 - **Public support and/or opposition**³
 - **Local workforce composition**²—e.g., for data-center-specific jobs and having a sufficient workforce to staff the data center.

We can either capture these factors directly where data exist or use quantitative adjustment factor(s) to holistically preference existing and popular new markets in siting.

¹Ashare, M. 2025. “Data Center Vacancy Rates Hit Record Low.” Utility Dive Brief. June 26, 2025. <https://www.ciodive.com/news/data-center-building-boom-capacity-crunch-rent-spikes-cbre/751669/>.

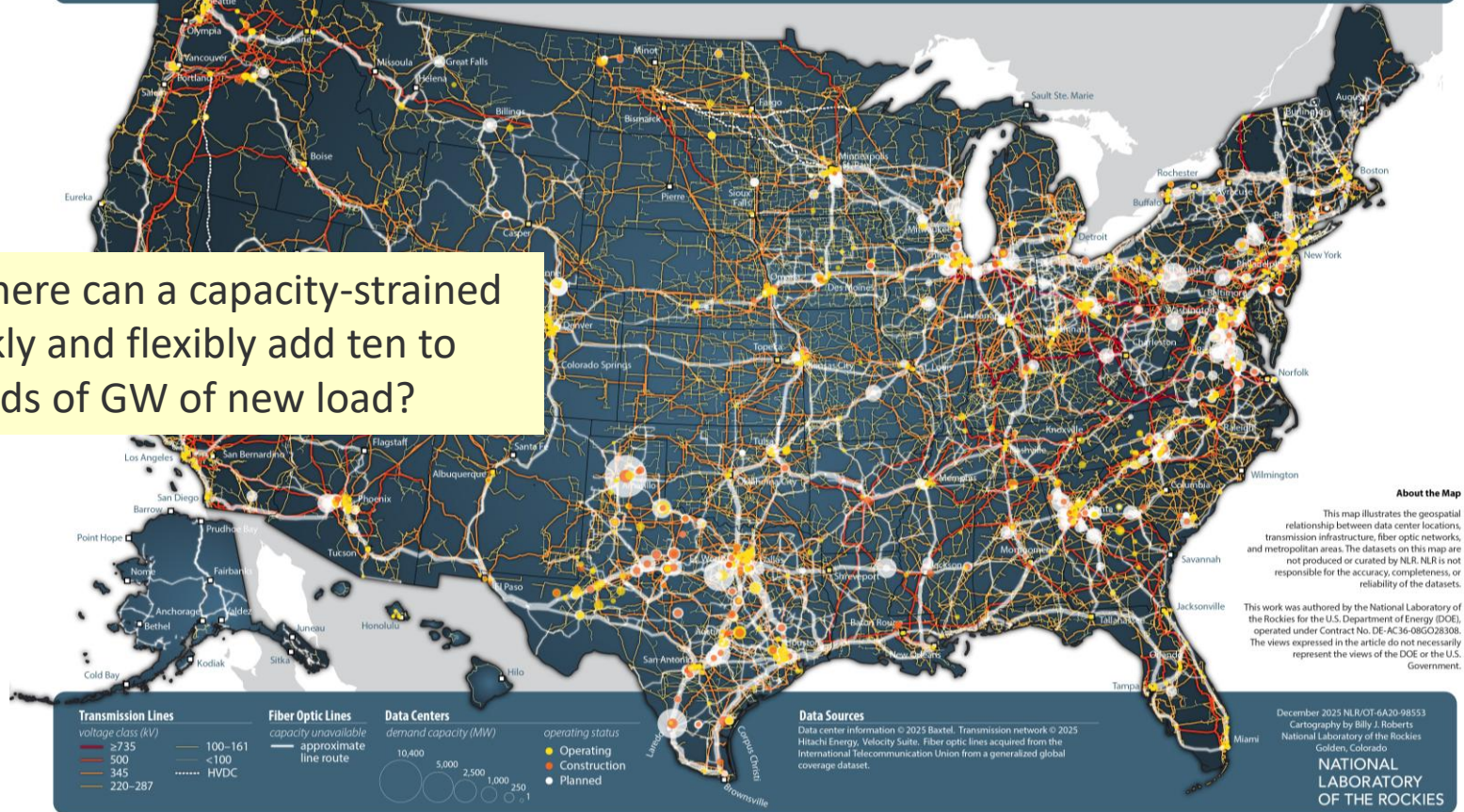
²Carey, M. 2025. “How Data Center Alley Is Changing Northern Virginia.” Oxford American. January 17, 2025. <https://oxfordamerican.org/oa-now/how-data-center-alley-is-changing-northern-virginia>.

³“\$64 Billion of Data Center Projects Have Been Blocked or Delayed Amid Local Opposition.” Data Center Watch. <https://www.datacenterwatch.org/report>; Murphy, R., and Feng, E. 2025. “Why More Residents Are Saying ‘No’ to AI Data Centers in Their Backyard.” NPR. Accessed September 2025. <https://www.npr.org/2025/07/17/nx-s1-5469933/virginia-data-centers-residents-saying-no>.

Mitigation Options for Near-Term Large Load Integration

Data Center Infrastructure in the United States, November 2025

How and where can a capacity-strained grid quickly and flexibly add ten to hundreds of GW of new load?



The “HAM” mitigation options: Hedge, Align, Match

Several approaches and/or mechanisms to potentially address growing large load demand have been floated in the literature or are emerging in real-world practice.

“HEDGE”: Tariffs



How? Structure tariffs to encourage demand flexibility or other investments.

Impact: Ensure large loads pay incremental costs.

Examples: Demand flexibility, storage, rates, and other tariff provisions

“ALIGN”: Streamlining Interconnection



How? Fast-track solutions to more efficiently use the grid and bring on new supply.

Impact: Align timelines for supply with large loads.

Examples: Grid-enhancing technologies (advanced conductors, advanced power flow control devices, dynamic line ratings, grid-forming batteries, etc.), provisional interconnection

“MATCH”: Co-location



How? Large loads physically locate and financially contract with proximate energy generation.

Impact: New large loads bring own matching energy supply.

Examples: On-site generation, front-of-meter arrangements, islanded arrangements

“Hedge”: Characterizing tariff design options

- Satchwell et al. (2025) note that “regulators and utilities are exploring tariff designs for large-load customers to **achieve desired policy and financial outcomes, provide reliable power to data centers, and mitigate risks and adverse impacts to other ratepayers.**”¹
 - Satchwell et al. also note **utilities have considerable latitude to design tariffs**, which broadly cover eligibility, contract duration, and energy sources.¹
- Electricity tariffs have multiple objectives but could affect costs of data center development and speed to power.
 - Creative/flexible provisions may improve speed to power. **Helpful questions to ask include:**
 - Who bears the financial risk of underutilized demand capacity? (Is it the data center developer? Other utility customers? Does this change over time?)
 - Under what conditions is capacity not firm (e.g., up-front demand flexibility or curtailable load arrangements)?

¹Satchwell, A., et al. 2025. “Electricity Rate Designs for Large Loads: Evolving Practices and Opportunities.” LBNL Technical Brief. January 2025.
https://eta-publications.lbl.gov/sites/default/files/2025-01/electricity_rate_designs_for_large_loads_evolving_practices_and_opportunities_final.pdf.

“Hedge”: Characterizing tariff design options

Mitigation Option	Mechanism/Theory of Change	Data Gaps	Quantifiable
Demand flexibility agreements	Demand response agreements with utilities (incentive-based) or proprietary software enabling energy shifting (e.g., Google’s Carbon-Aware Computing platform) ¹	Load data at finer geographic, temporal, and sub-sectoral resolutions (i.e., by data center type) to assess flexibility potential	Yes
Interruptible and/or non-firm service and non-capacity-backed load	Allow faster interconnection while allowing utilities to curtail or interrupt service if needed	Same as for demand flexibility—need for load profiles	Yes
Take-or-pay provisions	Contract provisions to enable risk sharing, cost recovery	Public, nationwide industrial and/or specific large load tariff rates database	Yes
Rates and rate structure	Incentivize peak load management, reduce financial or other utility risk exposure, create opportunities for price-based demand response	Public, nationwide industrial and/or specific large load tariff rates database	Yes
Demand charges, interconnection fees, and other aspects of rate structure	Incentivize peak load management, reduce financial or other utility risk exposure	Public, nationwide industrial and/or specific large load tariff rates database	Yes
Brownfield conversion projects	Utilize existing land to speed up otherwise lengthy permitting processes	Database of eligible brownfield conversion sites	Yes

¹Penrod, E. 2023. “Google Taps ‘Carbon-intelligent Computing Platform To Help Maintain Grid Reliability in Power Crises.” Utility Dive. November 7, 2023. National Laboratory of the Rockies | <https://www.utilitydive.com/news/google-carbon-intelligent-computing-platform-system-reliability-demand-response-grid-emergency/698958/>.

“Align”: Streamlining interconnection options ►►

Load Letter – Data Center

General Information

Data Center Address		Work Request Number (If available)
Meter Description (If multiple meters on same address)		
Electrician Name	Address	Phone ()
Engineering Consultant Name	Address	Phone ()
Owner Name	Address	Phone ()

Voltage at Meter ☐ 120/208 Volts ☐ 277/480 Volts ☐ Primary Voltage	Meter Location ☐ Inside ☐ Outside	Transformer(s) ☐ Owned by Utility ☐ Owned by Customer	If secondary is owned by Customer No. of Sets Size (kcmil) Type (Al or Cu)
--	---	---	---

Total Square Footage	Raised Floor Square Footage	☐ Check here to request Impedance for Arc Flash calculations
----------------------	-----------------------------	---

Backup Generation ☐ Break before Make/Open Transition (Preferred) ☐ Make before Break/Closed Transition (Company Approval required)	Amount of Capacity (MW) Required for Load Bank Testing/Commissioning
---	--

Electric Load (Excluding Motor Load)			
Interior Lighting _____ kW	Computers _____ kW	Power Factor	
Exterior Lighting _____ kW	Receptacles _____ kW		
Water Heating _____ kW	Electric Cooking _____ kW		
Heat Pump _____ kW	Chiller Plant _____ kW	UPS Losses	
Heat Pump Strip Heat _____ kW	Motor Control Center _____ kW		
Electric Heat _____ kW	Other _____ kW	Switchgear Size	
AC Data Processing Load _____ kW	Other _____ kW		
AC Not Including Data Processing _____ kW			
Total Requested Load _____ kW			

Example of a data center load request, Dominion Energy⁴

- Streamlining interconnection can pertain to both **process changes** and **options to speed up physical interconnections**.
 - Most fast-track interconnection processes are proposed by ISO/RTOs **on the supply side** (see Appendix).
 - Accurately forecasting load and assessing large load integration require a clear picture of where data centers are planned on the load side.
 - Load requests are not subject to “standardized transparency requirements.”¹
 - Load-side queues also receive distortionary “phantom requests.”²
 - Grid-enhancing technologies (GETs) comprise options to build out transmission capacity and/or exploit available headroom to create interconnection opportunities on the grid.³

¹“Phantom Data Centers Are Flooding the Load Queue.” Latitude Media. March 26, 2025. <https://www.latitudemedia.com/news/phantom-data-centers-are-flooding-the-load-queue/>.

²Martucci, B. 2025. “A Fraction of Proposed Data Centers Will Get Built. Utilities Are Wising Up.” May 15, 2025. <https://www.utilitydive.com/news/a-fraction-of-proposed-data-centers-will-get-built-utilities-are-wising-up/748214/>.

³U.S. Department of Energy Office of Electricity. 2023. “Grid-Enhancing Technologies: From Research and Development to Reality.” DOE Office of Electricity. November 13, 2023. https://www.energy.gov/oe/articles/grid-enhancing-technologies-research-and-development-reality-0?nrg_redirect=474378.

⁴“Load Letter – Data Center.” Dominion Energy. Accessed January 2026. <https://cdn-dominionenergy-prd-001.azureedge.net/-/media/content/large-business-services/pdfs/virginia/data-center-load-letter.pdf?rev=5af80441e2584283a96ae3f127f884a0>.

“Align”: Streamlining interconnection options ►►

Characterizing mitigation options that fall under “streamlining interconnection”

Mitigation Option	Mechanism/Theory of Change	Data Gaps	Quantifiable
Expedited or provisional interconnection process	Accelerate interconnection timelines by allowing for provisional interconnection, which requires something in exchange from the data center such as the utility’s right to curtail energy	Comprehensive demand-side interconnection data (often not public or aggregated in one place)	Maybe
Advanced conductors + grid-forming storage	Rapidly increase system capacity to enable faster generation- and load-side interconnection	No significant data gaps; Sienna ¹ and ReEDS™ ² can already model advanced conductors and other grid-enhancing technologies on the transmission system	Yes
Energy storage	Increase transmission capacity, alleviate congestion, demand flexibility	Estimated on-site and/or proximate storage resources and backup generation values by data center locations; finer resolution load data to enable modeling	Yes
Cluster- or batch-based study approach	Study proposed data center projects in a cluster rather than serially to speed up interconnection	Examples of cluster-based approaches for data centers	Maybe

¹For an overview of Sienna, see Barrows, C., Dalvi, S., Dhulipala, S., Doubleday, K., Henriquez Auba, R., Konar-Steenberg, G., Lara, J. D., Sanchez Perez, P., & Thom, D. (2024). Sienna Modeling Framework. <https://doi.org/10.2172/2376836>.

²For ReEDS model documentation, see Ho, Jonathan, Jonathon Becker, Maxwell Brown, Patrick Brown, Ilya Chernyakhovskiy, Stuart Cohen, Wesley Cole, et al. 2021. Regional Energy Deployment System (ReEDS) Model Documentation: Version 2020. Golden, CO: National Renewable Energy Laboratory. NREL/TP-6A20-78195.

<https://www.nrel.gov/docs/fy21osti/78195.pdf>.

“Match”: Co-location Arrangements



Co-location arrangements tend to fall into different “types,” as summarized below from emerging practices

Co-location Archetype	Description
Traditional on-site generation	Load connected to grid and behind-the-meter (BTM) generation powers it directly; both the grid and BTM generation can serve the data center load. Data center has the option to export net surplus electricity to the grid from on-site generation.
Co-location with shared interconnection (front-of-meter)	Load and generation share same point of interconnection (POI) or substation with the grid; data center has grid access. The data center primarily benefits from proximity to generation.
Isolated co-location (behind-the-meter)	Generation source directly powers the data center, which is not interconnected to the grid. Generator solely serves the data center.
Others	Microgrids with grid interconnections, utility-directed co-location (not a private developer partnership), direct physical line from generator/private wire power purchase agreement (PPA).

- **Co-location refers to** data centers receiving energy from large, proximate generation.
 - **Presumed benefits** include circumventing costly grid upgrades, allowing for additional flexibility, although this has raised stakeholder concerns around cost allocation and reliability.
 - **Currently there is no pathway recognized by the Federal Energy Regulatory Commission (FERC) for co-location.**
 - **There is significant, growing interest in** defining and exploring co-location arrangements for data centers.¹
- **Co-location options are evolving,** based on large load tariff proceedings and distinct regional tariffs.
- **There is a data gap** in terms of what co-location agreements are being struck and their *time to power*.

Options for Labs To Advance Large Loads Modeling

Opportunity for DOE national labs to fill gaps where transparent data and modeling are lacking

- Achieving improved model representation of data centers and improved speed to power requires better, quantified data.
- Several data gaps for quantifiable information about possible mitigation options point to a national lab role to fill these gaps.
- DOE labs can leverage their expertise in developing, curating, maintaining, and disseminating large datasets.



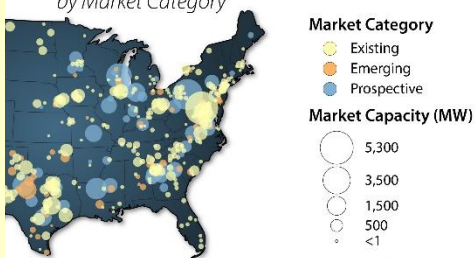
Photo by Werner Slocum, National Laboratory of the Rockies

Tracking metrics to reduce uncertainty in build-out needs

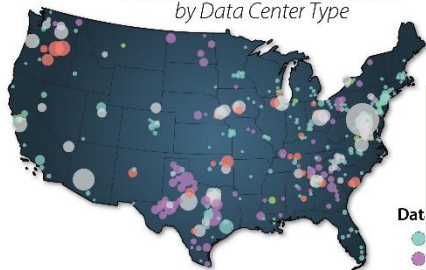
Data Center Markets, 2025

We see certain markets emerging... **Why?**
System headroom,
fiber optic lines, close
to generation?

All Data Center Markets
by Market Category



Existing & Emerging Markets
by Data Center Type



Prospective Markets
by Data Center Type

→ Answers indicate
what policymakers
could do to make their
area favorable



About the Data

Transmission network and data center information © 2025 Hitachi Energy.
Velocity Scale: Fiber optic lines were acquired from the International
Telecommunication Union from a generalized global coverage dataset.

August 2025
Cartography by Billy J. Roberts

- **Metrics to track uncertainty** could have a near-term impact in decision-making.
- **Regulators need independent info** to make real-time decisions.
- **Ideas for quantifiable components**
 - **Favorability:** Track MW by market and large load type; output metrics on which areas/criteria have seemed “favorable” in recent year(s)
 - **Grid costs:** Bulk system cost projection scenarios
 - **Hedging:** Track in-use large load tariffs
 - **Aligning:** Track prevalence of co-location arrangements and types
 - **Matching:** Track interconnection processes
 - **Mitigating:** Longer-term options, system headroom, and/or GETs

Ideas for national labs to pursue data development and modeling

Based on data gaps, to understand mitigation options and modeling opportunities at the national labs, **we present three ideas for data development:**

1

Develop **representative load profiles** for different data center types

2

Track metrics on **co-location arrangements** to improve modeling representation of co-location options

3

Gather and track detailed data on **in-use large load tariffs**

1. Highly resolved, representative load profiles and projections by data center type

- **Clear need for higher-resolution load data for different data center types¹** to evaluate demand flexibility and study bulk system cost projection scenarios
- **Ongoing national lab efforts** to produce hourly industrial load profiles at the county level
- **Different estimation approaches** with pros/cons for each
- **Example input data:**
 - **Consumption:** Aggregate consumption, hourly or sub-hourly power usage effectiveness (PUE), sub-process load factors
 - **Data center types and characteristics:** Standard facility characteristics, equipment and IT stacks, IT nameplate power capacity ratings, rack counts, servers by class, etc.
 - **Point locations and footprints:** Geospatial point locations from Baxtel, facility footprints where available
 - **Region and/or climate zone:** Ambient weather for modeling of cooling processes—for example, from National Oceanic and Atmospheric Administration (NOAA) or National Solar Radiation Database (NSRDB) datasets
 - **Rates and tariff structure:** To allow for possibility of price- or behavior-based demand

¹For example, EPRI has stated that “there is limited empirical data for the temporal pattern of actual power use for different DC configurations...Different types of data centers exhibit a variety of load patterns based on component configurations and system specifications.” “Powering Data Centers.” 2024. EPRI.

<https://www.epri.com/research/products/000000003002031198.>)

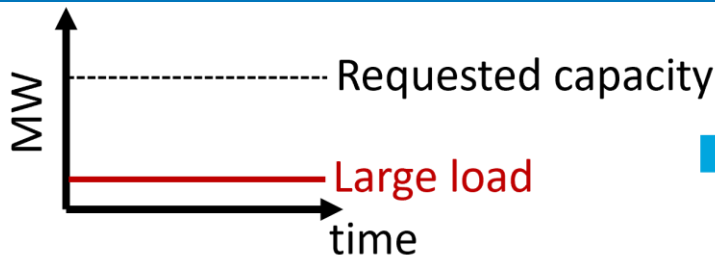
Multiple valid approaches to developing “canonical” large load profiles

Approach	Pros	Cons
Meter data	Empirical	Under non-disclosure arrangement, vintaging issues for applying to new AI workloads
Utilization data or typical utilization rates (e.g., utilities, regulatory agency, consulting assumptions)	Available in literature and regulatory filings	Often generic assumptions
Bottom-up model training/inference efficiencies	Empirical, some available data in public domain	Based on limited public and/or non-representative (e.g., high-performance computing) data
Job scheduling data (e.g., from Ravel, HPC)	Understand from first principles how much flexibility there is in the compute side before dealing with electricity	Understanding application programming interfaces, data gaps may take staff time
Nodal load data with known large loads (e.g., from VelocitySuite, OASIS matched against Baxtel)	Empirical	Matching could be difficult, data could lag/be delayed
ComStock™/physics-based simulation	Has been done successfully for other building types	Potentially labor intensive and not yet started
Mapping/geospatial/image recognition (e.g., identifying new substations at data center locations)	Empirical	Requires sophisticated tooling applied to up-to-date geospatial information
Multiply power densities by facility footprints or estimated square footage	Empirical, simplistic	May not capture enough temporal variation; no clear advantage over current profiles

Three archetypes of potential load profiles and related challenges for data center buildout

1. Low-utilization:

Large load underutilizes requested capacity

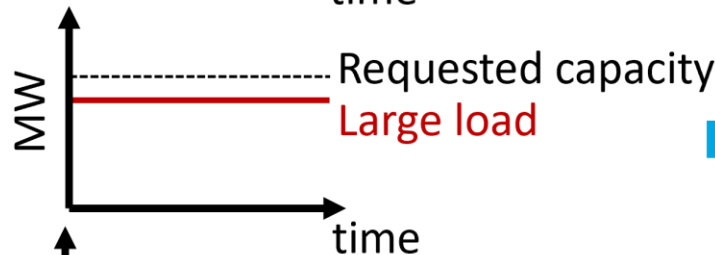


Challenge: Who bears cost for oversized capacity?



2. Flat:

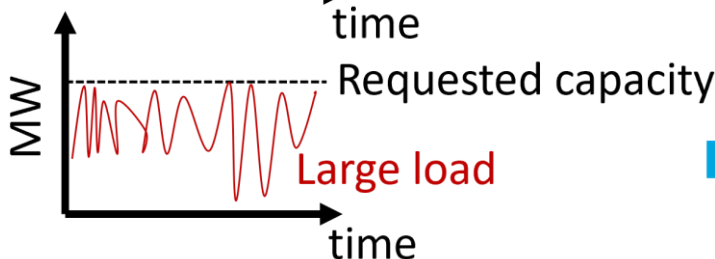
Consistently high utilization large load



Challenge: Maintaining resource adequacy

3. Fast-ramping:

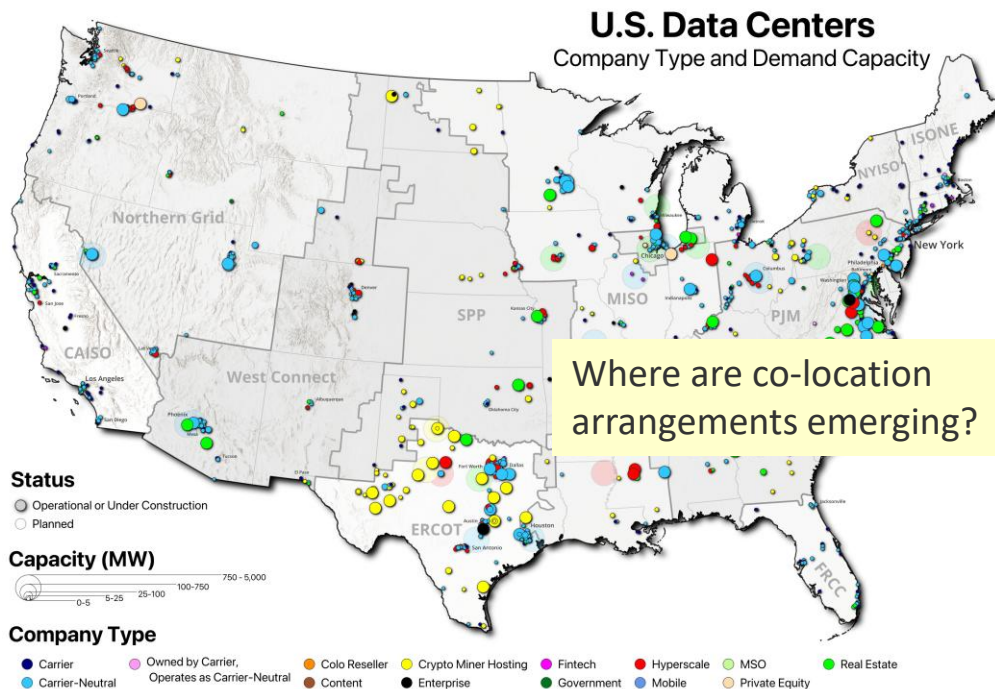
Swings in power consumption



Challenge: Need for batteries or other resources to firm load

Solutions for cost-effectively serving new load likely vary by archetype.

2. Tracking co-location arrangements



Source: U.S. Data Center Sites. Baxtel, 2025. <https://baxtel.com>. Accessed June 16th, 2025.

- **Tracking co-location arrangements** could improve modeling and allow for analysis of options to improve time to power.
- **Next steps** would entail defining a set of metrics to track for existing/new builds based on Sienna modeling capabilities.
 - Data center location, size, and other details
 - Generation source details
 - Single/shared POI
 - Behind-the-meter/Front-of-meter
 - Operational vs. planned.

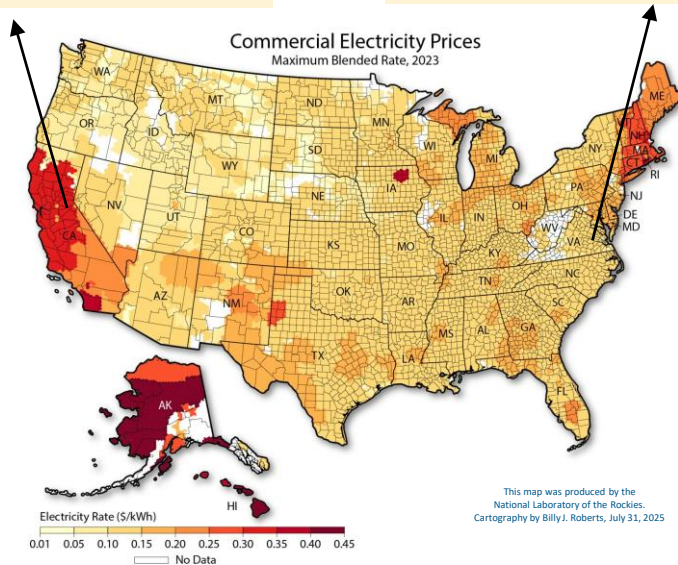
Source: U.S. Data Center Sites. 2025. Baxtel. Accessed June 16, 2025. <https://baxtel.com>.

3. Tracking in-use and emerging tariffs can provide insight into near-term large load development

Guidance is evolving. Goal for Labs could be to track list of core provisions and enrollment of customers where possible (e.g., through Utility Rate Database).

California SB57 requires establishment of specific rate structures for data centers.

Virginia SCC approves new rate class for large customers



For clarity, blended utility rates by utility service territory from EIA Form 861 have been summarized to a uniform 25km grid, and the highest blended rate is shown where multiple territories overlap. Rate data has been overlaid with state and county boundaries for reference.

Source: Billy Roberts, NLR 2025.

Database of Emerging Large-Load Tariffs (DELTA)

SEPA's DELTA tracks information such as utility, tariff name, year approved, minimum load, and other select tariff characteristics

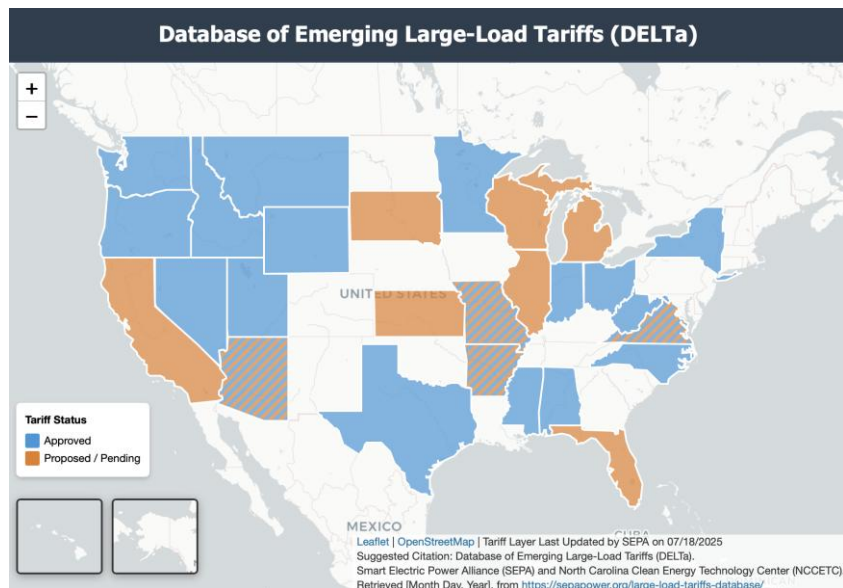
Can we track other quantitative criteria (e.g., rates, applicable demand charges, fixed/downpayment costs)?

Tariff Status
Approved
Proposed / Pending

MEXICO
Leaflet | OpenStreetMap | Tariff Layer Last Updated by SEPA on 07/18/2025
Suggested Citation: Database of Emerging Large-Load Tariffs (DELTA).
Smart Electric Power Alliance (SEPA) and North Carolina Clean Energy Technology Center (NCCETC).
Retrieved [Month Day, Year], from <https://sepapower.org/large-load-tariffs-database/>

Smart Electric Power Alliance (SEPA). Database of Emerging Large-Load Tariffs (DELTA). Accessed August 2025. <https://sepapower.org/large-load-tariffs-database/>.

Tracking in-use and emerging large load tariffs



Source: SEPA DELTA.

- Potential exists for collaboration based upon existing datasets (EPRI, SEPA DELTA)^{1,2} to produce an **NLR-vetted, public database of emerging large load tariffs.**
- **Database could include elements** such as rates, demand flexibility, interruptible vs. firm service requirements, and cost allocation.
- Large load tariffs could comprise **another layer of NLR's rates database (USURDB)**³ and be combined with geospatial data on data center locations.
- Next steps would likely entail discussing with potential collaborators, understanding web-scraping tools to scrape ongoing regulatory proceedings and approved tariff filings.

¹ Retail Flexible Load Tariff Sample Dataset. Electric Power Research Institute. <https://www.epri.com/research/products/3002033645>.

² Database of Emerging Large Load Tariffs (DELTA). Smart Electric Power Alliance. <https://sepapower.org/large-load-tariffs-database/>.

³ Jimmy-Schmitt, Daniel, and Jay Huggins. *Utility Rate Database (URDB)*. NLR, June 1, 2010. Distributed by Open Energy Data Initiative (OEDI). <https://data.openei.org/submissions/5>.

Appendix

ISO/RTO fast-track interconnection processes

ISO/RTO	Interconnection Process Description	Load/Gen	Sources
CAISO	Fast-track study process for generation customers <5 MW requesting energy-only deliverability	Gen	CAISO
ERCOT	No specific fast-track process for large load interconnections; generally uses a “connect-and-manage” approach but recently passed legislation on other aspects of large loads	None	TX SB6
MISO	FERC-approved Expedited Resource Addition Study (ERAS) on the supply side	Gen	MISO
PJM	Issued decisional letter on a Critical Issue Fast Path (CIFP) Initiative that includes tracks for expedited and provisional interconnection of large loads		PJM , Utility Dive
SPP	FERC-approved framework for Conditional High Impact Large Load (CHILL) and High Impact Large Load Generation Assessment (HILLGA) processes for fast-track large load interconnections; FERC-approved Expedited Resource Adequacy Study (ERAS) on the supply side	Both	SPP , SPP
ISO-NE	No fast-track interconnection plans or proposals for large loads announced	None	N/A
NYISO	No fast-track interconnection plans or proposals for large loads announced	None	N/A

Disclaimer: These developments were captured at the time of publication and may have changed or evolved.
 ISO = independent system operator; RTO = regional transmission operator.

State-level model large load tariff actions

State	Large Load Tariff Actions or Discussions	Source
PA	PA Public Utility Commission (PUC) has published a tentative order for a model large load tariff and solicited commentary on a range of tariff design issues.	PA PUC , PA PUC
OH	Ohio PUC approved a data center tariff for AEP Ohio , an electric utility, following proceedings at the PUC.	PUCO , PUCO
VA	Virginia State Corporation Commission approved new rate class for large customers > 25 MW	VA SCC
NC	North Carolina is exploring large load issues through holding technical meetings.	Canary Media , Utility Dive
TX	Texas SB6 introduced several provisions , including mandatory emergency curtailment for large loads and new rules regarding interconnection, cost-sharing, and co-location.	TX SB6
CA	California SB57 requires establishment of specific rate structures for data centers.	CA SB57
OR	Oregon HB 3546 created a new rate class for large loads >20 MW and requires large loads to pay for transmission costs and a minimum amount of energy across contracts.	OR 3546
IN	Indiana Regulatory Commission ruled on a large load tariff joint settlement that requires data centers to make long-term financial commitments, among other requirements.	Renewable Energy World



Thank You

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